Software Engineering HSC 2025 - sample full mark response

Question 16 (9 marks)

A weather app is to be designed to display real-time updates such as:

- rain commencing within the next 10 minutes
- weather event warnings and warning cancellations
- location specific weather conditions.

(a) Compare the use of a progressive web app (PWA) with an interactive website, for delivering this weather app. (3 marks)

Sample

A progressive web app, designed to operate as a native application on a device, would offer features such as offline access through caching, however for this whether app, the live updates require live up to date real time information updates. The PWA would however be able to push notafications such as weather warnings. By using a interactive website, hosted by a sever and accessed directly from a web broweser, can display this live updated data. Both of these methods can encorperate location specific conditions.

(b) Outline TWO ways, other than the use of a PWA, to minimise the load time of this weather app. (3 marks)

Sample

Two ways to minimise the load time of a weather app, would be data caching, where comminly used data is temporarily stored to be reused, and data compressions, where the data size, and resolution of data is selected, appropriaite to its application.

(c) Describe ONE way that the 'privacy by design' approach could be implemented in this weather app. (3 marks)

Sample

The 'privacy by design' approach could be implemented in the weather app by ensuring privacy is an area of focus from the beginning of the development lifecycle, with requirements definitions and boundaries. For example, when defining the features and characteristics of the app's ability to show location specific weather conditions, privacy should be carefully considered; the storage of user locations should be minimsed where possible to reduce potential liability and increase data security and integrity, which in turn builds customer trust. It should be considered how users will be notified and clearly asked for their permission to use their location data.

Question 18 (3 marks)

A streaming service has exclusive rights to broadcast a live event each Sunday afternoon.

Explain how the streaming service can use metadata and data mining to prevent technical difficulties during the broadcast of this weekly event.

Sample

By using metadata and data mining, streaming service can collect the website traffic and peak user time of the website or the app on a specific time. This allows the streaming service to predict the high user using time and plan the broadcast of this weekly event into a different time where it is less user on the streaming service to prevent technical difficulties. Steaming service can also use the collected data from data mining to see what is the most common problems that most user are dealing with so they can fix the problem and improve their steaming platform before the event occur.

Question 19 (3 marks)

In an organisation, temporary passwords are automatically generated for new user accounts. Each password is created from the first three letters of the username followed by ‘123’.

For example, the username ‘NESA’ will have the password ‘NES123’.

When a user logs in for the first time, they are asked for their username and password. They receive a welcome message or an error message, depending on whether the password entered follows the rules of new user accounts.

Write a Python program that asks a new user for their username and password, and then checks whether the password rule has been followed.

Sample

Student input

```
# Input code
username = input("What is your username: ")
username = list(username)
password = username[0] + username[1] + username[2] + "123"
pw_check = input("What is your password: ")
if (pw_check == password):
    print("Welcome!")
else:
    print("Error! Your password did not follow the rules")
```

Student output

```
# Output
What is your username: NESA
What is your password: NES123
Welcome!
```

Question 20 (3 marks)

Describe how session management can affect the efficiency of code execution for the user.

Sample

Session management, when used effectively, can greatly increase the efficiency of code execution for a user. This is because session management allows for sessions to be saved up to a certain amount of time so that if a user were to go back to a previous site, there session and progress would have been saved on that site and it would be quicker to load. Furthermore, session management can automatically close inactive sessions after a certain amount of time which allows for more performance from a users computer on other sites and sessions

Question 21 (4 marks)

Below is the Issues table for a technical support company. FixTime, in minutes, is the time taken to complete each job.

JobID	Issue	Response	FixTime	Category	Status
0001	Cannot connect to the internet	Clear cache	22	Software	Pending
0002	Installation failure	Reinstall the printer drivers	76	Hardware	Pending
0004	Forgot password	Reset password	97	Software	Resolved
0005	Installation failure	Follow the installation guide	110	Hardware	Escalated
0009	Blue screen error	Run a system diagnostic test	120	Software	Pending
0011	Installation failure	Verify email settings	45	Software	Escalated

Construct a SQL query that displays:

- **the number of pending software jobs**
- **the number of pending hardware jobs**
- **the number of escalated software jobs**
- **the number of escalated hardware jobs**

List the results in order from the highest to lowest, based on the number of jobs.

Sample 1

Query

```
1  SELECT Status, Category, COUNT(*) AS Number
2  FROM Issues
3  WHERE ((Category = 'Software') OR (Category = 'Hardware'))
4  AND ((Status = 'Pending') OR (Status = 'Escalated'))
5  GROUP BY Status, Category
6  ORDER BY COUNT(*) DESC
```

Status	Category	Number
Pending	Software	2
Escalated	Hardware	1
Escalated	Software	1
Pending	Hardware	1

Sample 2

Query

```
1  SELECT Category, Status, COUNT(*) AS "Count"FROM Issues
2  WHERE Status = "Pending" OR Status = "Escalated"
3  GROUP BY Status, Category
4  ORDER BY Count Desc
5
```

Category	Status	Count
Software	Pending	2
Hardware	Escalated	1
Software	Escalated	1
Hardware	Pending	1

Question 22 (3 marks)

An animal classification tool is to be developed. The classes to be used are shown below.

Mammal
Colour
Size
Birth type
Number of teeth

Reptile
Colour
Size
Scale type
Tongue type

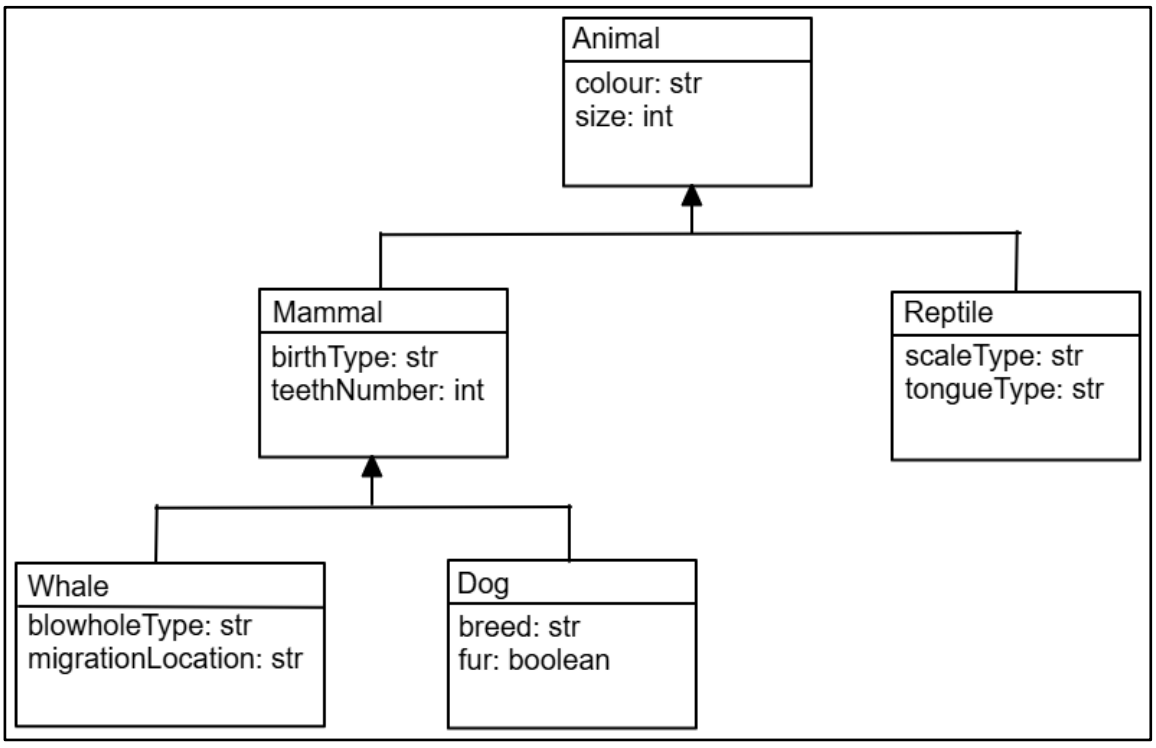
Whale
Colour
Size
Birth type
Number of teeth
Blowhole type
Migration location

Dog
Colour
Size
Birth type
Number of teeth
Breed
Fur

Construct a class diagram that shows inheritance and removes redundant attributes for each class. Additional classes may be included if required.

You do NOT need to show the nature of the relationships between the classes.

Sample



Question 23 (9 marks)

A food delivery service offers prepackaged meals tailored to different dietary preferences, for example, vegetarian and gluten-free. It has developed a website for customers to place orders. The website is ready for testing and feedback.

The website includes:

- a menu that dynamically updates based on meal availability
- user accounts where customers can track deliveries and manage subscriptions
- a backend that processes orders and payments, updates stock and communicates with the delivery team.

(a) Outline potential social and ethical issues associated with the use of this website. (3 marks).

Sample

Use of this website would require users to input sensitive information - location data, for the purposes of food delivery, and financial details for the purposes of payment and subscription. Thus, a potential ethical issue would be the leaking of these sensitive personal details in the event of a data breach, exposing critical information to malicious attackers. Furthermore, if the integrity of customer dietary preference data were to be compromised, customers may be delivered food in violation of their dietary preferences, creating a significant social issues with potential health consequences.

(b) Explain why server-side web programming would be needed to develop this website.

Support your answer with examples from the website. (3 marks)

Sample

Sever-side web programming is the implementation of code that functions between the database and website, allowing for clients to securely make requests with sensitive information and store their data without fear for a lack of confidentiality. For example, the user needs to securely transfer their bank information and process orders to the website server, preventing potential malcious intenders from injecting javascript code that could manipulate and stealing unauthorised information. Additonally, the server-side web programming can update and change the stored data in the database to future new requests, preventing any errors regarding buying an item that may not be in stock.

Question 24 (5 marks)

A construction company introduces a safety monitoring system that uses artificial intelligence. Cameras and wearable sensors track worker movements and environmental conditions. The system detects unsafe behaviours through artificial intelligence, such as:

- standing too close to operating machinery
- entering restricted zones
- failure to wear personal protective equipment (PPE).

Alerts are sent in real-time to site supervisors, and long-term data is stored for further analysis.

Discuss the possible impact of this monitoring system on workers.

Sample 1

The possible impacts of the monitoring system can positively boost safety and productivity. Through the use of artifical intelligence, workers are incentivised to avoid unsafe behaviours, preventing potential injury and mistake throughout the work-zone. Furthermore the wearable sensors allows site supervisors to supervise workers that may act unaccordingly and underproductive, further showing midigated risk of injury and reduced chance of workers failing to comply to work regulations. Long-term data allows supervisors to monitor each worker, ensuring safety while monitoring productivity, efficiency at completing tasks, and creates a data base analysising users, allowing supervisors to arrange the workers in order of safest to unsafe, potentially causing shifts in job positions and workers being let off.

Sample 2

The introduction of this system should lead to increased safety of workers as they are able better understand safety risks of the workplace and be reported of any unsafe actions they are performing without concious decision. It will additionally discourge risky workplace behavour as workers can be held accountable for their actions by supervisors. If someone is identied as a repeat offender due to the long term storage of data they can be removed from the workplace, increasing personal and colleague safety. However it can also lead to a disregard of safety by workers and they think that the cameras will detect the risks and they can simply do their job, introducing oversight in potentially hazardous environments. This could also be viewed as an infringement of personal privacy and some workers may feel uncomfortable with the adoption of this additional security measure leading to misuse or risky behaviour.

Question 25 (5 marks)

An airline introduces a virtual personal assistant as an initial support contact point for cancelled flights, missed connections, or lost luggage. The virtual personal assistant uses a neural network to:

- analyse support queries to direct customers to the appropriate department
- recommend appropriate tone and wording to be used by the human customer support representative
- assess data such as support turnaround time, customer ratings, and repeat contacts.

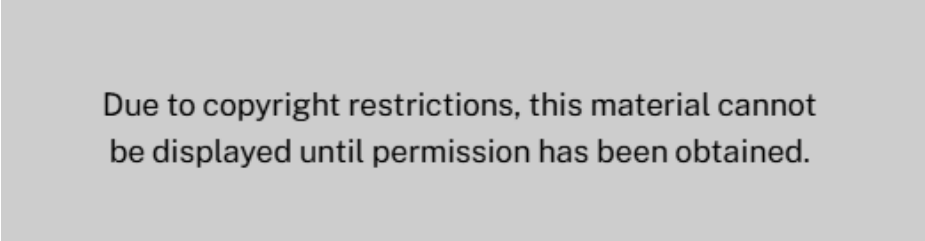
Explain how the neural network supports the virtual personal assistant. Include ONE positive and ONE negative impact of using the virtual personal assistant.

Sample 1

Neural networks are machine learning algorithms inspired by the human brains that have an input layer, hidden layer and output layer. The hidden layer has interconnected nodes that form as linear regression models, that each have a certain weighting which determines the importance of an output. Hence these neural networks support the virtual assistant as it aids in making decisions depending on the given scenario, based on the weighting of each node. A positive impact is that it reduces human error and efficiently routes customers to appropriate departments based on a set of input scenarios. This also helps save time and need for human intervention. If the customer has entered that they lost luggage, the neural network will process this and output to the customer the appropriate department. A negative however, is that this neural network may only be fitted to a range of certain scenarios or outcomes, and if a customer has a very specific or unusual need, the virtual assistant may be unable to help as it is untrained in for that specific input. So for this case, human input and adaptability would be better.

Question 27 (6 marks)

Watch the video and then answer the question.



Explain how human bias could affect the usefulness of artificial intelligence (AI) solutions.

Support your answer with reference to the video.

Sample 1

Human bias can significantly affect the usefulness of articifially intelligent solutions, particularly those based on machine learning models. Machine learning is a subset of AI, and involves a model learning from data to analyse patterns and trends, and make predictions without being explicitly programmed to do so. Artificial intelligence is the ability of machines to process information as humans do, and reflect this in actions, decisions and behaviour.

However, as stated in the video, a machine learning model is not necessarily neutral just because it learns from data, as learns solutions by finding patterns in data, and the accuracy and reliability is dependant on the quality of the training dataset.

Bias is when a model makes overly simple assumptions about a relationship, and is reflected in machine learning models when datasets contain human bias. The video referenced how bias is reflected in human actions and decisions indirectly. For example, when asked to draw a show, many people chose to draw a similar style, which would influence how a machine learning model learns what a shoe is and how to identify it.

An AI solution is only useful if the data provided for it to learn from is diverse and accurately represents the problem that needs to be solved. Therefore, to make an AI solution more useful, databases should composed of data from a diverse range of sources that does not discriminate one group over another. Databases should be audited by multiple people, from multiple perspectives, to ensure there is no human bias that may influence the decisions made by the model, thus affecting its usefulness.

Sample 2

Despite artificial intelligence being based on indifferent data, human bias can still affect the usefulness of artificial intelligence solutions due to the inability to escape from our own human biases. This video provides examples supporting this conclusion, demonstrating the negative impacts and reinforced biases that can occur do to human bias interfering with the training of AI solutions.

If AI models are trained through structured learning using labelled data, issues can arise rom the choice of labelled data selected. The video demonstrates this through the example of only using labelled data of the same shoe, meaning the alogirthm is only able to recognise one type of shoe. This can cause inaccurate classifications and occurs directly as a result of human bias. Not only does this affect theusefulness and accuracy of AI solutions, but it can also have potentially harmful outcomes in other contexts. For example, if this issue occured in a medical context, where human bias caused specific symptoms of a disease to be left out of model training, the patient could be given an inaccurate diagnosis.

Human bias can also cause artificial intelligence solutions to gain prejudice against certain groups, another problem caused through human bias implemented in model training. The video demonstrates this, showing that if AI solutions are only trained on past male physicians, the algorithm will end up with a bias towards men. This can create significant social and ethical issues as it causes the misrepresentation of groups and reinforcement of prejudice.

Ultimately, the usefulness of AI solutions can directly be negatively impacted as a result of human bias implemented in the training stage - potentially causing inaccurate classifications and bias against certain groups.